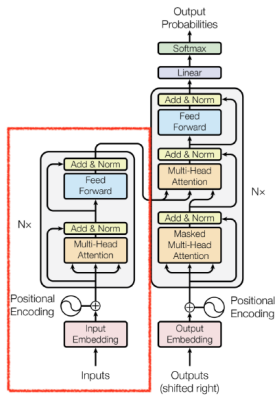


Lecture 3: Transformers and encoder-based models in Social Science

Miriam Hurtado Bodell

Recap transformers/encoders



Encoders (transformers) in the social sciences

How can we use encoder part of transformer models?

Discovery

- In contrast to WE models, individual vectors and their relations may not be the main interest – but similarly to WE you can **study relations between words/sentences to explore cultural associations** (which can handle polysemy better!)

Measurement

- The main task to use encoder (so far) is **classification** of the presence of relevant sociologically relevant concepts in a text

Discovery: encoders (transformers) in the social sciences

What does vectors from (some different hidden layer(s) in) the encoder in transformer models represent?

- Technical answer: Numerical representation of a word's position in a vector space, taking context into consideration
- Sociological answer: ?

Discovery: encoders (transformers) in the social sciences

What does vectors from (some different hidden layer(s) in) the encoder in transformer models represent?

- Technical answer: Numerical representation of a word's position in a vector space, taking context into consideration
- Sociological answer: The **contextualized** meaning of a word

→ meaning is again relational, but also contextual!

Discovery: encoders (transformers) in the social sciences

No consensus on which layers best capture the specific contextualized semantic meaning of a word

- Input: static embeddings
- First layer: probably contains very limited contextual info
- Last layer: may be overfitted for the training task
- Second to last layer may be best to use, or a combination of the last couple of layers

Measurement: encoders (transformers) in the social sciences

What type of sociologically relevant concept can we use encoders to measure? For example;

- Ideology or frames

“Protecting our borders and upholding the Second Amendment are essential to preserving our **American** values and ensuring the safety and security of our citizens.”

"Ensuring that every **American** has access to affordable healthcare and quality education is fundamental to our nation's progress and prosperity"

Example using contextualized embeddings for **measurement**: own work in progress, Hurtado Bodell & Goldberg (2026?)

Aim: Show that cultural change depends on networks of meaning

Cultural change is often studied by looking at how attitudes/meanings change around individual concepts. But meaning emerges through relations between concepts → there are constraints which concepts will diverge

Data: One important cultural change = polarization. Politics (speeches from floor)

Example using contextualized embeddings for **measurement**: own work in progress, Hurtado Bodell & Goldberg (2026?)

Method: Extract embeddings from a RoBERTa model, and measure in-group cohesion relative to between-group separation (silhouette score)

Use attention score to create “semantic networks” →

(i) Study if the divergence of concept w at time t moves in the direction of it's neighbors N_w at time $t - 1$ (semantic adaption),

(ii) Study if divergence of concept w at time t , depends on the divergence level of it's neighbors N_w at time $t - 1$ (semantic clustering)

→ signal that cultural change/disagreement is constrained by the structure of the semantic space

Example using contextualized embeddings for **measurement**: own work in progress, Hurtado Bodell & Goldberg (2026?)

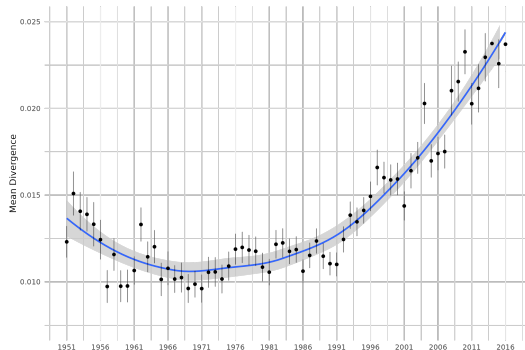


Table 1: Comparison of baseline models without and with word fixed effects.

	Eq. 3 (1)	Eq. 5 (2)	Eq. 3 (3)	Eq. 5 (4)
Constant	0.0108*** (0.0002)	0.0046*** (0.0002)		
$\tilde{D}_{Nw,t-1}$	0.5059*** (0.0521)		0.2804*** (0.0302)	
$G_{w,t-1}$		0.4961*** (0.0319)		0.6934*** (0.0270)
Word FE	No	No	Yes	Yes
Observations	249,423	249,423	249,423	249,423
Adjusted R ²	0.01440	0.24266	0.30616	0.34075

Clustered standard errors in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Measurement: encoders in the social sciences: why is fine-tuning important for social scientists?

Pre-trained encoder is however mostly used for measurement task by **fine-tuning** them = target task-specific output layer is put on top of the pretraining architecture

Pre-trained models *may* encode information about language/culture

- Models can learn supervised classification tasks with less labeled training data
- Researcher doesn't have to (or pay others to) annotate as much training data

Example fine-tuning for **measurement**: Le Mens et al. (2023)

Aim: Measure **typicality** of literary genres using BERT

What is typicality? Related to the literature on **categorization** (large lit. within for example organization science) = (mental) process of deciding in X an instance of concept Y; e.g. is *Avatar* a sci-fi movie?

Commonly thought about as a binary classification task; yes or no but **typicality** measure the **degree** to which X can be seen as an instance of concept Y

Example fine-tuning for **measurement**: Le Mens et al. (2023)

Data: Descriptions of books from 36 different genres (partly focus on Mystery and Romance) + humanly annotated prediction set

Method: Feed text to pre-trained BERT model. Use BERT embeddings as input in neural net to classify what genre books belong to; add softmax layer to get probability $p_c \rightarrow \tau_c = \log \frac{p_c}{p(c)}$

Compare τ_c with human annotator asked to answer how typical a book is to a genre (0-100)

Example fine-tuning for **measurement**:

Typicality measure	Correlations between model-based typicalities and human typicality ratings			Model components		Similarity between text document and concept
	Romance and Non-Romance books	Romance books	Non-Romance books	Language representation	Categorization	
BERT fine-tuned / categorization probability (baseline BERT typicality)	0.86	0.54	0.72	BERT	Probabilistic binary	Typicality (Eq. [4])
BERT fine-tuned / correlation with prototype	0.85	0.56	0.70	BERT	Probabilistic binary	Cosine
BERT pre-trained / categorization probability	0.70	0.40	0.55	BERT	Probabilistic binary	Typicality (Eq. [4])
BERT pre-trained / correlation with prototype	0.07	0.11	0.04	BERT	None	Cosine
BERT fine-tuned / categorization probability 36	0.85	0.51	0.67	BERT	Probabilistic 36 categories	Typicality (Eq. [4])
GloVe fine-tuned / categorization probability	0.77	0.53	0.54	GloVe	Probabilistic binary	Typicality (Eq. [4])
GloVe fine-tuned / correlation with prototype	0.59	0.27	0.47	GloVe	Probabilistic binary	Cosine
GloVe pre-trained / categorization probability	0.76	0.45	0.56	GloVe	Probabilistic binary	Typicality (Eq. [4])
GloVe pre-trained / correlation with prototype	0.30	0.30	0.22	GloVe	None	Cosine
Word frequencies / categorization probability	0.62	0.34	0.42	BoW term frequencies	Probabilistic binary	Typicality (Eq. [4])
TF-IDF / categorization probability	0.76	0.48	0.54	BoW TF-IDF	Probabilistic binary	Typicality (Eq. [4])

Example fine-tuning for **measurement/inference**: Vicinanza et al. (2023)

Aim: Answer where prescient ideas come from; the core or the periphery of different domains? And how does it relate to success?

Prescient ideas = first challenge conventions, then become widely accepted (e.g. civil rights). (1) novel & (2) future-looking

Data: Politics (speeches from floor), Law (court rulings), and Business (Quarterly earnings calls)

Example fine-tuning for **measurement/inference**: Vicinanza et al. (2023)

Method: Fine-tune one BERT model per time unit (year or presidential term)

Calculate **contextual novelty** (CN)= word-level perplexity in a sentence; how “surprised” is the model from $t = 0$ and $t = 1$ to see a word in a sentence

Calculate **sentence prescience** = $\frac{CN_{t0} - CN_{t1}}{CN_{t0}}$

Example fine-tuning for **measurement/inference**: Vicinanza et al. (2023)

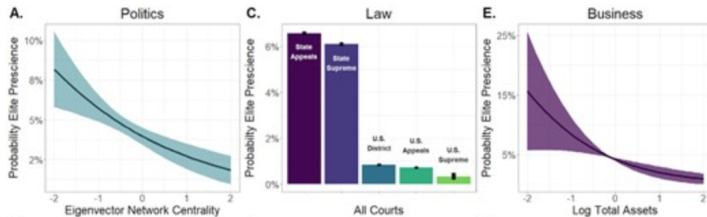


Figure: Prescient ideas come from the periphery in all domains

Example fine-tuning for **measurement/inference**: Vicinanza et al. (2023)

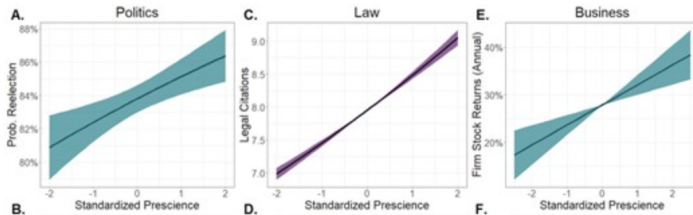


Figure: Prescient actors are more successful

What encoder to use?

Depend on **the task** you want them to perform and **your data**

- Different models are available for different languages
- Did they train on data similar to yours?
- How long are your texts; different models can deal with different context sizes (e.g. BERT/RoBERTa 512 tokens)

No general guidelines, study model performance across tasks that are relevant to your study!

*) Wankmüller, S. (2022); Timoneda, J. C., Vallejo Vera, S. (2024)

Validation

The main validation technique is to compare with human annotation or human responses

- Annotate random sample and compare with classification output
- Use found labeled data (e.g. reviews)
- Conduct a survey and compare with the classification output

Validation

Encoder models should not replace social scientists, but augment them!
(Do et al. 2024)

- The quality of measurement (classification) still depends on having high-quality labels to fine-tune/validate your model
- Who conduct the labeling matter; the more expert knowledge the better the model performance in less time

Limitations

Pre-trained models may “bleed” unwanted information from pre-training to measurement/inference task

- Your validation data is part of training data
- Measurement/inference about subgroups/individuals will be influenced by others in training data

Summary

Encoder-only transformers has shown great ability in both simple and complex measurement tasks

But their ability still depends on human annotation; the better the training/validation labels the better the model performance

Important to consider how training data may “spill” unwanted information into your inference

Thank you!

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